PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Association of occupational sitting with cardiovascular outcomes and cardiometabolic risk factors: A systematic review with a sex/gender-sensitive perspective
AUTHORS	Reichel, Kathrin; Prigge, Michaela; Latza, Ute; Kurth, Tobias; Backe, Eva- Maria

VERSION 1 – REVIEW

REVIEWER	Timoteo, Ana Teresa Ctr Hosp Lisboa Cent
REVIEW RETURNED	14-Feb-2021

GENERAL COMMENTS	The authors performed a systematic review of longitudinal studies to update the evidence on the association of occupational sitting / sedentarism with cardiovascular outcomes and cardiometabolic risk factors. The main outcomes studied were all-cause mortality, cardiovascular mortality, cardiovascular heart diseases, BMI and obesity and cardiometabolic risk markers. Data screening and extraction was performed by three independent reviewers according to PRISMA guidelines. Methodology is described comprehensively and follows strict guidelines and recommendations for systematic reviews, and includes a PRISMA Checklist They found marked heterogeneity between studies, regarding measurements, as well as cardiometabolic risk factors and cardiovascular morbidity and mortality. Of the 27 studies selected, nine found association with the primary outcome in the overall sample, five found an association in different subgroups, and in eleven, sex-differences were found. They concluded that the evidence is limited, but there seems to be an association with several cardiometabolic outcomes. A detailed description of strengths and limitations is also included and in my opinion they were all correctly identified. After this systematic review, I agree with the authors that results remain inconclusive. It is of particular importance to mention (as the authors do) that these results are different from previous studies on overall sedentary behavior in the general population, where this association has been conclusively demonstrated, suggesting some differences with occupational sitting that should be identified. There is a marked heterogeneity of exposure measurement and for that reason, future research with a prospective design and a well standardized and reliable measurement of occupation sitting is warranted. I only suggest that the term “sex” and “gender” should be used in only one format

REVIEWER	Rahimlou, Mehran Ahvaz Jundishapur University of Medical Sciences
REVIEW RETURNED	23-Apr-2021

GENERAL COMMENTS	The systematic review and meta-analysis by Reichel and coworkers focuses on the Association of occupational sitting with cardiovascular outcomes and cardiometabolic risk factors: A systematic review with a sex/gender-sensitive perspective. This is a very interesting topic and can answer many questions in this area. However, some major concerns have affected the results of this study. 1. As the authors of this article point out, one of the major concerns associated with higher heterogeneity of measurements of occupational sitting and the wide variety of measured outcomes. 2. Also, the difference in the work of the participants in the study and the design of the study is one of the important factors affecting the results. 3. It was better in the conclusion section; the conclusion was written based on high quality studies. 4. Also in the discussion section, it would have been better to write more necessary explanations based on the results of higher quality studies.

VERSION 1 – AUTHOR RESPONSE

Response to Reviewer #1:
Dear Dr. Ana Teresa Timoteo

We appreciate your careful reading and would like to say thank you for the comments. In conclusion you wrote:

“I only suggest that the term “sex” and “gender” should be used in only one format.” to which we respond as follows:

We recognize that there are different discussion cultures in different countries and understand your suggestion to use the terms “sex” and “gender” in one format only. We like to refer and connect to Doull et. al. [15] as part of the Cochrane Sex/Gender Methods Group suggesting to use the term “sex/gender” in recognition of the theoretical and empirical entanglement of the two terms “sex” and “gender” (see p. 4, 3rd paragraph of introduction). As this comment is a suggestion to consider and the other reviewers did not comment on the format, we would prefer to choose the recommended format “sex/gender”. Further, as an explanation in the introduction of the manuscript we added:

“As shown above in this context, sex-based biological factors and gendered social factors are interrelated. Recognizing the entanglement of the terms “sex” and “gender”, hereinafter we use the term “sex/gender” recommended by the Cochrane Sex/Gender Methods Group [15].”

However, if you feel more comfortable with the term “gender” using only one format and the editors follow your suggestion, we can alter the last sentence of paragraph as follows to: "…hereinafter in this paper we use the term “gender”." and use the term gender throughout the text.

Best regards from the author team

Response to Reviewer #2:
Dear Dr. Mehran Rahimlou

We appreciate your careful reading and would like to say thank you for the comments! You expressed the following major concerns that have affected the results of this study to which we respond as follows:

“1. As the authors of this article point out, one of the major concerns associated with higher heterogeneity of measurements of occupational sitting and the wide variety of measured outcomes.“

To 1. You highlighted that, as we had pointed out, one of the major concerns associated with higher heterogeneity of measurements of occupational sitting is the wide variety of measured outcomes.

We totally agree on this point. In order to highlight this, we had included details in the results section on p. 17 under “Heterogeneous assessments of exposure to occupational sitting”, systematically summarized in Table 2, explained on p. 19 last paragraph (“Study heterogeneity”) and p. 19ff. Additionally, we included a 2nd paragraph in the discussion section on p. 24:

“The informative value of this review seem to be mainly restricted by the extreme heterogeneity of both measurements and reference categories of occupational sitting as well as the variety of measured outcomes considered in the studies included. Further, the spectrum of study participants’ work was broad ranging from population samples to single occupations, e.g. one high quality study included nurses only[38, 39].”

“2. Also, the difference in the work of the participants in the study and the design of the study is one of the important factors affecting the results.”

To 2. You underline that the difference in the work of the participants in the study and the design of the included studies are one of the important factors affecting the results.
We absolutely agree on this point. We described those differences in the results section on p. 7 under “study selection” and “study setting and study characteristics” and summarized this closing on p. 19 under “Study heterogeneity”. We included explanatory notes in the manuscript to the discussion section on p. 24, see #1 above and #4 of your peer-review further down.

“3. It was better in the conclusion section; the conclusion was written based on high quality studies.”

To 3. You commented to the conclusion section that it would be better if the conclusion was based on high quality studies.
Yes, we agree and focused the conclusion mainly on the results of the included high quality studies. Thus, the results section from the abstract was altered as follows:

“… The review included 27 high or acceptable quality publications. Of the eight high quality publications from seven cohorts, three cohort studies found significant associations of occupational sitting with primary outcomes. Additionally, one study described an association with the secondary outcome. Another high quality publication found an association between occupational sitting and IHD in a subgroup already at risk due to hypertension. For sex/gender analysis, 11 of the 27 high and acceptable quality publications reported sex-stratified results. Five of these found sex-differences.”

“4. Also in the discussion section, it would have been better to write more necessary explanations based on the results of higher quality studies.”

To 4. You stated that in the discussion section, it would have been better to write more necessary explanations based on the results of higher quality studies.
Thank you for this comment regarding the discussion section. We agree with this suggestion, revised the discussion and added information regarding the results of the high quality studies in five paragraphs of the discussion section:

“In this systematic review of studies considering the effect of occupational sitting on cardiometabolic outcomes, we identified eight high quality publications that presented results from seven cohort studies. Three cohort studies[38, 39, 47, 43] found significant associations of occupational sitting with the primary outcomes. The study by Chau et al. (2015) reported no association with the primary outcome CVD mortality but with the secondary outcome all-cause mortality. In a sub analysis of the study by Allesoe et al.[38], occupational sitting was associated with IHD in the subgroup already at risk due to hypertension[39]. The strength of the association identified in the included high-quality studies was weak to moderate.
The results based on the high quality studies are similar to the results when additional 19 publications with at least acceptable quality are considered. Out of all 27 publications included, nine showed associations of occupational sitting with cardiometabolic outcomes for the overall sample. Further, the 27 publications allow to answer questions regarding subgroups. Risk estimates seem to be higher in at-risk subgroups (e.g., Allesoe et al.[39]). Four of six publications that considered occupational sitting together with a lack of sufficient LTPA reported a higher risk of cardiometabolic outcomes for the combination of both exposures. …”
”… Further, the spectrum of study participants’ work was broad ranging from population samples to single occupations, e.g. one high quality study included nurses only[38],[39]…”
“…. Underestimation of risk due to misclassification of exposure remains an issue. The reference category in four of the seven high quality studies (eight publications) comparing different intensities of OPA was heavy or moderate manual work. Four of the seven high quality studies considering different duration of sitting time also used different categories. …. Additionally, six of the eight high quality publications (and most of all publications included) examined the general working population, without details on occupational groups or contexts. …”
“…Nonetheless, at least half of the high quality studies suggested an association with the primary or secondary outcome. …”

Best regards from the author team